

1. LENGTH

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', the subject 'PYTHON PROGRAMMING', and a user profile for 'Dara Kusumanjali'. The left sidebar shows 'QUESTIONS 0 / 20 completed' and a dropdown menu with 'Length' selected. The main area is titled 'PROGRAM EDITOR' and contains the following Python code:

```
1 t = tuple(input("Enter a tuple (comma-separated values): ").split(','))
2 print("Length of the tuple is:", len(t))
```

Below the code editor are 'Run' and 'Save' buttons. On the right, the 'INPUT' field contains the value '5'.

2.

3. REVERSE OF A TUPLE

The screenshot shows the SIMATS Online Programming Lab interface. The header is the same as the previous screenshot. The left sidebar shows 'QUESTIONS 2 / 20 completed' and a dropdown menu with 'Reverse' selected. The main area is titled 'PROGRAM EDITOR' and contains the following Python code:

```
1 t = tuple(input("Enter tuple elements separated by space: ").split())
2 print("Reversed tuple:", t[::-1])
```

Below the code editor are 'Run' and 'Save' buttons. On the right, the 'INPUT' field contains the values '1 2 3 4'.

4. MIN MAX

The screenshot shows the SIMATS Online Programming Lab interface. The header is the same as the previous screenshots. The left sidebar shows 'QUESTIONS 2 / 20 completed' and a dropdown menu with 'Max-Min' selected. The main area is titled 'PROGRAM EDITOR' and contains the following Python code:

```
1 t = tuple(map(int, input("Enter tuple elements separated by space: ").split()))
2
3 max_val = max(t)
4 min_val = min(t)
5
6 print("Maximum element:", max_val)
7 print("Minimum element:", min_val)
```

Below the code editor are 'Run' and 'Save' buttons. On the right, the 'INPUT' field contains the values '10 20 30 40'.

CONCATE

The screenshot shows the SIMATS Online Programming Lab interface. The header is the same as the previous screenshots. The left sidebar shows 'QUESTIONS 3 / 20 completed' and a dropdown menu with 'Concatenate' selected. The main area is titled 'PROGRAM EDITOR' and contains the following Python code:

```
1 t1 = (1, 2, 3)
2 t2 = (4, 5, 6)
3
4 result = t1 + t2
5 print("Concatenated tuple:", result)
```

Below the code editor are 'Run' and 'Save' buttons. On the right, the 'INPUT' field contains the text 'Your INPUT goes here!'.

CONVERSION OF TUPLE AND LIST



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

Dara Kusumanjali
 192425398LRD

QUESTIONS 4 / 20 completed
 Convert
CONVERT
 Write a Python program to convert a tuple to a list.

PROGRAM EDITOR
 Python Run Save

```

1 t = (1, 2, 3, 4)
2 l = list(t)
3
4 print("Tuple:", t)
5 print("List:", l)

```

INPUT
 Your INPUT goes here!

FIND INDEX



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

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QUESTIONS 5 / 20 completed
 Index
INDEX
 Write a Python program to find the index of a given element in a given tuple.

PROGRAM EDITOR
 Python Run Save

```

1 t = (10, 20, 30, 40, 50)
2 x = int(input("Enter the element to find: "))
3
4 if x in t:
5     print("Index of", x, "is:", t.index(x))
6 else:
7     print("Element not found in the tuple.")

```

INPUT
 20

CHECK ELEMENT EXISTS IN TUPLE



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

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QUESTIONS 6 / 20 completed
 Exist
EXIST
 Write a Python program to check if a given element exists in a given tuple.

PROGRAM EDITOR
 Python Run Save

```

1 tup = tuple(map(int, input("Enter the tuple elements (space-separated): ").split()))
2
3 x = int(input("Enter the element to check: "))
4
5 if x in tup:
6     print("Element exists in the tuple.")
7 else:
8     print("Element does not exist in the tuple.")

```

INPUT
 10 20 30 40
 20

CONVER TUPLES TO LISTS



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

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QUESTIONS 7 / 20 completed
 Convert
CONVERT
 Write a Python program to convert a list of tuples to a list of lists.

PROGRAM EDITOR
 Python Run Save

```

1 def convert_tuples_to_lists(tuple_list):
2     return [list(t) for t in tuple_list]
3
4 input_list = [(1, 2), (3, 4), (5, 6)]
5 output_list = convert_tuples_to_lists(input_list)
6
7 print(output_list)

```

INPUT
 Your INPUT goes here!

SORTING THE TUPLES



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

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QUESTIONS 8 / 20 completed
 Sort
SORT
 Write a Python program to sort a given tuple in ascending order.

PROGRAM EDITOR
 Python Run Save

```

1 t = tuple(map(int, input("Enter elements of the tuple separated by space: ").split()))
2 sorted_t = tuple(sorted(t))
3 print("Sorted tuple (ascending):", sorted_t)

```

INPUT
 10 40 30 20

SUM OF ELEMENTS IN A TUPLE



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 PYTHON PROGRAMMING

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QUESTIONS 9 / 20 completed
 Sum
SUM
 Write a Python program to find the sum of all elements in a given tuple.

PROGRAM EDITOR
 Python Run Save

```

1 t = (1, 2, 3, 4, 5)
2 total = sum(t)
3 print("Sum of all elements in the tuple is:", total)

```

INPUT
 Your INPUT goes here!

CONCATENATE 2 DICTIONARIES



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

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QUESTIONS 11 / 20 completed
 Concatenate
CONCATENATE
 Write a Python program to concatenate two dictionaries.

PROGRAM EDITOR
 Python Run Save

```

1 dict1 = {"a": 1, "b": 2}
2 dict2 = {"c": 3, "d": 4}
3 result = **dict1, **dict2
4 print("Concatenated Dictionary:", result)

```

INPUT
 Your INPUT goes here!

EXIST KEY IN DICTIONARY



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 PYTHON PROGRAMMING

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QUESTIONS 12 / 20 completed
 Exist-key
EXIST -KEY
 Write a Python program to check if a given key exists in a given dictionary.

PROGRAM EDITOR
 Python Run Save

```

1 my_dict = {'name': 'Kusumanjali', 'age': 20, 'course': 'Python'}
2
3 key_to_check = input("Enter the key to check: ")
4
5 if key_to_check in my_dict:
6     print("Key exists in the dictionary.")
7 else:
8     print("Key does not exist in the dictionary.")

```

INPUT
 age

EXSISTS VALUE

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PYTHON PROGRAMMING

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QUESTIONS 13 / 20 completed
Exist - Value
EXIST - VALUE
Write a Python program to check if a given value exists in a given dictionary.

Python Run Save

```

1 value = input("Enter the value to check: ")
2
3 d = {
4     "a": 10,
5     "b": 20,
6     "c": 30
7 }
8
9 found = value in d.values()
10 print("YES" if found else "NO")

```

INPUT
20

REMOVE THE GIVEN KEY FROM THE DICTIONARY

SIMATS Online Programming Lab
PYTHON PROGRAMMING

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QUESTIONS 14 / 20 completed
RM Key
RM KEY
Write a Python program to remove a given key from a given dictionary.

Python Run Save

```

1 def remove_key(d, key):
2     if key in d:
3         del d[key]
4     return d
5 my_dict = {"name": "KUSUM", "age": 20, "city": "Chennai"}
6 key_to_remove = "age"
7
8 print("Before:", my_dict)
9 result = remove_key(my_dict, key_to_remove)
10 print("After: ", result)

```

INPUT
{"name": "KUSUM",
"age": 20, "city":
"Chennai"}
age

MAX MIN VALUES

SIMATS Online Programming Lab
PYTHON PROGRAMMING

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QUESTIONS 15 / 20 completed
Max-Min
MAX-MIN
Write a Python program to find the maximum and minimum values in a given dictionary.

Python Run Save

```

1 d = {1: 10, 2: 5, 3: 20, 4: 2}
2
3 max_value = max(d.values())
4 min_value = min(d.values())
5
6 print("Maximum Values:", max_value)
7 print("Minimum Value:", min_value)

```

INPUT
Your INPUT goes here!

MAX AND MIN VALUES IN THE DICTIONARY

SIMATS Online Programming Lab
PYTHON PROGRAMMING

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QUESTIONS 16 / 20 completed
Max-Min
MAX-MIN
Write a Python program to find the keys with the maximum and minimum values in a given dictionary.

Python Run Save

```

1 def max_min_keys(d):
2     max_key = min_key = None
3
4     for key, value in d.items():
5         if max_key is None or value > d[max_key]:
6             max_key = key
7         if min_key is None or value < d[min_key]:
8             min_key = key
9
10    return max_key, min_key
11
12 d = { "a": 10, "b": 5, "c": 20, "d": 3 }
13
14 max_key, min_key = max_min_keys(d)
15 print("Maximum key:", max_key)
16 print("Minimum key:", min_key)

```

INPUT
Your INPUT goes here!

FREQUENCY OF THE ELEMENTS GIVEN IN DICTIONARY



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

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QUESTIONS 17 / 20 completed
 Frequency
FREQUENCY
 Write a Python program to find the frequency of all elements in a given dictionary.

PROGRAM EDITOR
 Python
 Run Save


```

1 def frequency_of_elements(d):
2     freq = {}
3     for key, value in d.items():
4         freq[value] = freq.get(value, 0) + 1
5     return freq
6
7 input_dict = {"a": 10, "b": 20, "c": 10, "d": 30, "e": 20}
8 result = frequency_of_elements(input_dict)
9 print("Frequency of elements:", result)

```

INPUT
 Your INPUT goes here!

SORT A GIVEN DICTIONARY BY VALUES



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

Dara Kusumanjali
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QUESTIONS 18 / 20 completed
 Sort
SORT
 Write a Python program to sort a given dictionary by its values.

PROGRAM EDITOR
 Python
 Run Save


```

1 d = {"apple": 5, "banana": 2, "cherry": 7, "date": 1}
2
3 sorted_dict = dict(sorted(d.items(), key=lambda item: item[1]))
4
5 print("Sorted dictionary:", sorted_dict)

```

INPUT
 Your INPUT goes here!

CONVERT DICTIONARY TO TUPLES



SIMATS Online Programming Lab
 PYTHON PROGRAMMING

Dara Kusumanjali
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QUESTIONS 19 / 20 completed
 Convert
CONVERT
 Write a Python program to convert a dictionary to a list of tuples.

PROGRAM EDITOR
 Python
 Run Save


```

1 d = {"a": 1, "b": 2, "c": 3}
2
3 list_of_tuples = list(d.items())
4
5 print(list_of_tuples)

```

INPUT
 Your INPUT goes here!